MedAgent Clinical Decision Support

Case ID: case_599ad77208 | **Status:** EDIT | **Physician:** Dr. Alex Vance | **Date:** 2026-10-15 20:20 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 72, Sex: Female
Notes & History: 72yo female with type 2 diabetes and hypertension presenting with progressive lethargy, nausea, oliguria for 24 hours. Currently taking lisinopril 20mg and spironolactone 25mg daily. Denies chest pain, denies hemoptysis, denies dyspnea.

Clinical Summary / Impression

Clinical evaluation indicates features consistent with Acute Kidney Injury, characterized pathophysiologically by abrupt decline in glomerular filtration rate leading to retention of nitrogenous wastes (azotemia), dysregulation of fluid balance, and impaired electrolyte and acid-base excretion. Key supporting clinical indicators include oliguria, Acute Kidney Injury / Renal Impairment, Hyperkalaemia, while secondary differential considerations include Severe Hyperkalemia, Metabolic Acidosis. Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

[URGENT CLINICAL ALERT] RED FLAGS IDENTIFIED:
• Critical hyperkalemia with lethal arrhythmia risk. Urgent ECG and stabilization required.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Acute Kidney Injury**	**HIGH**	oliguria, Acute Kidney Injury / Renal Impairment, Hyperkalaemia, Metabolic Acidosis	S_AcuteK_1, S_AcuteK_2
2	**Severe Hyperkalemia**	**MODERATE**	Hyperkalaemia, Acute Kidney Injury / Renal Impairment, Metabolic Acidosis	S2, S3
3	**Metabolic Acidosis**	**MODERATE**	Metabolic Acidosis, Acute Kidney Injury / Renal Impairment	S6, S_Chroni_1
4	**Hyponatremia**	**MODERATE**	lethargy, Hyponatraemia	S1
5	**Chronic Kidney Disease**	**MODERATE**	Acute Kidney Injury / Renal Impairment, Hyperkalaemia	S_Chroni_1, S_Chroni_2

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
Potassium	**6.4**	mEq/L	3.5 – 5	**CRITICAL HIGH**
Serum Creatinine	**3.8**	mg/dL	0.5 – 1.1	**HIGH**
BUN	**64**	mg/dL	7 – 20	**HIGH**
Sodium	**132**	mEq/L	135 – 145	**LOW**
Bicarbonate / CO2	**15**	mEq/L	22 – 29	**LOW**

Inferred Diagnoses from Laboratory Analysis:
• **[CRITICAL] Severe Hyperkalemia** (Likelihood: HIGH): Serum potassium elevated to 6.40 mEq/L, crossing the safety threshold (3.5–5.0 mEq/L) and posing acute cardiotoxicity risk. *Mechanism:* Dangerous elevation of extracellular potassium lowering myocardial resting membrane threshold, impairing cardiac repolarization, and predisposing to fatal ventricular arrhythmias or asystole.
• **[SEVERE] Acute Kidney Injury / Renal Impairment** (Likelihood: HIGH): Serum creatinine elevated to 3.80 mg/dL, demonstrating renal parenchymal filtration failure. *Mechanism:* Abrupt decline in glomerular filtration rate leading to retention of urea, creatinine, and nitrogenous metabolic waste products with disruption of fluid, electrolyte, and acid-base homeostasis.
• **[MODERATE] Metabolic Acidosis** (Likelihood: HIGH): Low serum bicarbonate (15.0 mEq/L). *Mechanism:* Pathological accumulation of unmeasured fixed organic acids (lactic acid, ketoacids, uremic toxins) consuming serum bicarbonate buffers.
• **[MODERATE] Hyponatremia** (Likelihood: HIGH): Serum sodium depressed to 132.0 mEq/L (reference interval 135–145 mEq/L). *Mechanism:* Excess of total body water relative to sodium content causing extracellular hypo-osmolality, driving water shift into intracellular spaces and predisposing to cerebral edema.

Synthesized Clinical Patterns:
• **[URGENT] Renal Impairment:** Elevated nitrogenous wastes and/or reduced filtration rate indicating acute or chronic renal dysfunction.
• **[URGENT] Hyperkalaemia:** Elevated serum potassium conferring risk of fatal cardiac arrhythmias.
• **[SOON] Hyponatraemia:** Subnormal serum sodium requiring cautious osmolality evaluation and rate-controlled correction.

• **[SOON] Metabolic Acidosis:** Decreased serum bicarbonate / total CO2 indicating systemic acid accumulation or bicarbonate loss.

Suggested Diagnostic Workup

• Renal ultrasound to exclude hydronephrosis
• Urine electrolytes and fractional excretion of sodium
• Discontinue nephrotoxic agents
• Immediate 12-lead ECG for peaked T waves/conduction blocks
• Calculate serum anion gap [Na - (Cl + HCO3)]
• Serum and urine osmolality
• Spot urine albumin-to-creatinine ratio

Physician Attestation: I, Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **EDIT**.
Reviewer Notes: No specific notes recorded.
Signature: _____ **Date:** 2026-10-15 20:20 UTC